Computational Logic

Predicate Logic - the language

Jakub Szymanik

- Week 1 (14.09): Intro (try to read Logicomix after)
- Week 2 (21.09): How to Win a Million Dollars with Logic? Part 1:The language and what it means. (Study Sections 1.1, 1.3, 1.4.1-1.4.2 of the book)
- Week 3: (28.09) How to win a million dollars with logic? Part 2: Proof - and why it matches meaning. (Study 1.2, 1.4.3-1.4.4)
- Week 4: (05.10) How to win a million dollars with logic? Part 3: Normal forms and solvers. Lab, BY laptops (Study 1.5, 1.6)
- Homework 1.
- Week 5: (12.10) FO and non-computability (Study 2.1-2.2)
- Week 6: (19.10) FO and non-computability (Study 2.3-2.4)
- Week 7: (26.10) FO and non-computability (Study 2.5) + Trial intermediate exam.
- Homework 2.
- Week 8: (02.11) Expressiveness, Definability, and Generalized Quantifier Theory (Study 2.6-2.7, and the book chapter on Moodle)
- Week 9: (09.11) Modal Logic and Agency 1 (Study 5.2-5.3)
- Week 10: (16.11) Modal Logic and Agency 2 (Study 5.4-5.5)
- Week 11: (23.11) Modal Logic and Agency 3.
- Homework 3.
- Week 12: (30.11) Q&A + Trial final exam.

Who got a CNF converter running? Did you try to blow it up? What did you get at n = 11?

Last week

- We discussed Horn formulas, for which SAT is easy.
- You implemented one of the two algorithms from §1.5.
- There is homework due this Friday.
- Everything in chapter 1 was decidable. Expensive — 2^n, NP-complete — but a machine could always, eventually, answer. From today, that stops being true.

Complex reasoning patterns

- Every student is younger than some instructor. Anna is a student. Bob is the only instructor. Therefore, Anna is younger than Bob.
- In propositional logic, this argument is p, q $\vdash$ r. Our proof system says it is invalid — and our soundness theorem says there is no proof. Both are correct. The problem is not the proof system.
- The language cannot see inside the sentences.

A syllogistic reasoning

Some zookeepers are pacifists.
No pacifists are troglodytes.

Does it follow that some zookeepers are not troglodytes?

FOUR SENTENCES — AND WHAT EACH ONE FORCES ON THE LANGUAGE

SAY THIS	AND YOU NEED
Every student is younger than some instructor $\forall x\,(S(x) \rightarrow \exists y\,(I(y) \wedge Y(x,y)))$	**nested quantifiers** *an ∃ inside the scope of a ∀*
Not all birds can fly $\neg\,\forall x\,(B(x) \rightarrow F(x))$	**scope of negation** *outside the quantifier, not inside it*
Every child is younger than its **mom** $\forall x\,(C(x) \rightarrow Y(x, \mathbf{m(x)}))$	**a function symbol** *every child has exactly one — a predicate would not say that*
Every man loves some woman $\forall x\,(M(x) \rightarrow \exists y\,(W(y) \wedge L(x,y)))$ $\exists y\,(W(y) \wedge \forall x\,(M(x) \rightarrow L(x,y)))$	**a decision English never made** *two formulas, one sentence — which did you mean?*

Predicate Logic

Elementary Logic, First-order logic

It’s an extension of PL with:

- Predicates
- Variables/constants
- Quantifiers

Predicate Logic as a formal language

Terms

Definition 2.1 Terms are defined as follows.

- Any variable is a term.
- If $c \in \mathcal{F}$ is a nullary function, then c is a term.
- If $t_1, t_2, \ldots, t_n$ are terms and $f \in \mathcal{F}$ has arity $n > 0$, then $f(t_1, t_2, \ldots, t_n)$ is a term.
- Nothing else is a term.

Predicate Logic as a formal language

Formulas

Definition 2.3 We define the set of formulas over $(\mathcal{F}, \mathcal{P})$ inductively, using the already defined set of terms over $\mathcal{F}$:

- If $P \in \mathcal{P}$ is a predicate symbol of arity $n \geq 1$, and if $t_1, t_2, \ldots, t_n$ are terms over $\mathcal{F}$, then $P(t_1, t_2, \ldots, t_n)$ is a formula.
- If ϕ is a formula, then so is $(\neg\phi)$.
- If ϕ and ψ are formulas, then so are $(\phi \wedge \psi)$, $(\phi \vee \psi)$ and $(\phi \rightarrow \psi)$.
- If ϕ is a formula and x is a variable, then $(\forall x\, \phi)$ and $(\exists x\, \phi)$ are formulas.
- Nothing else is a formula.

What is the difference between a function and a relation?

- Every child is younger than its mom.
- with a function m $\forall x\,(C(x) \to Y(x, m(x)))$
- with a relation M $\forall x\,(C(x) \to \exists y\,(M(y,x) \wedge Y(x,y)))$
- To make M behave like m you have to add it yourself:
- $\forall x\,\exists y\,M(y,x)$ — at least one $\forall x \forall y \forall z\,((M(y,x) \wedge M(z,x)) \to y = z)$ — at most one.
- A function symbol has those two built in. That is what "function" means.
- terms — m(x) is a thing; Definition 2.1.
- formulas — M(y,x) is a claim; Definition 2.3.
- So you can write m(m(x)) — my mother's mother — and you can never write M(M(y,x)). Terms are what you are allowed to substitute. That is why we keep function symbols.

$$\forall x\,((P(x) \to Q(x)) \wedge S(x, y)).$$

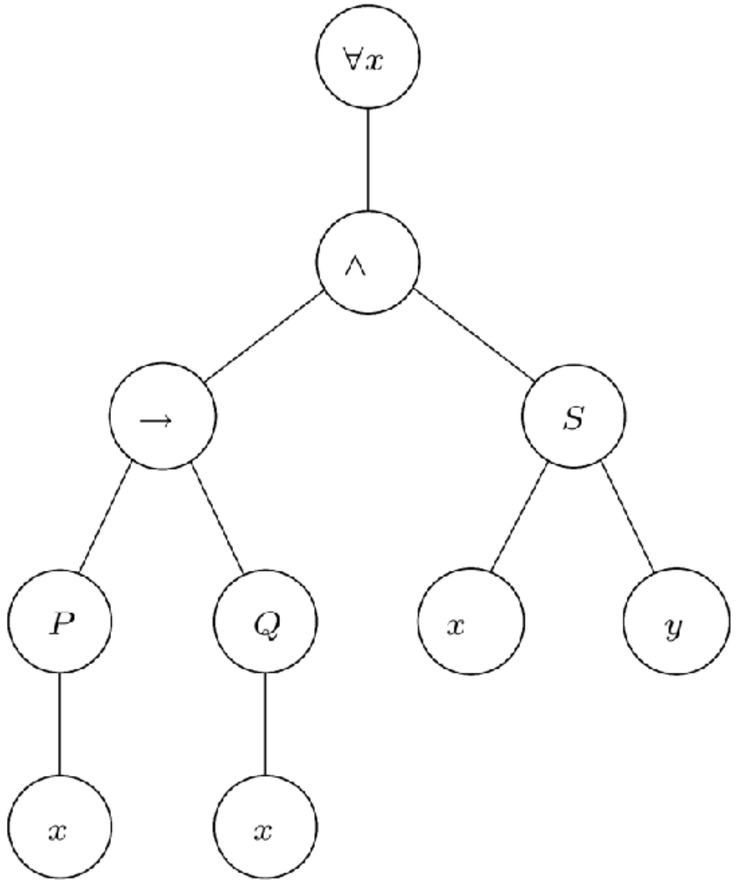
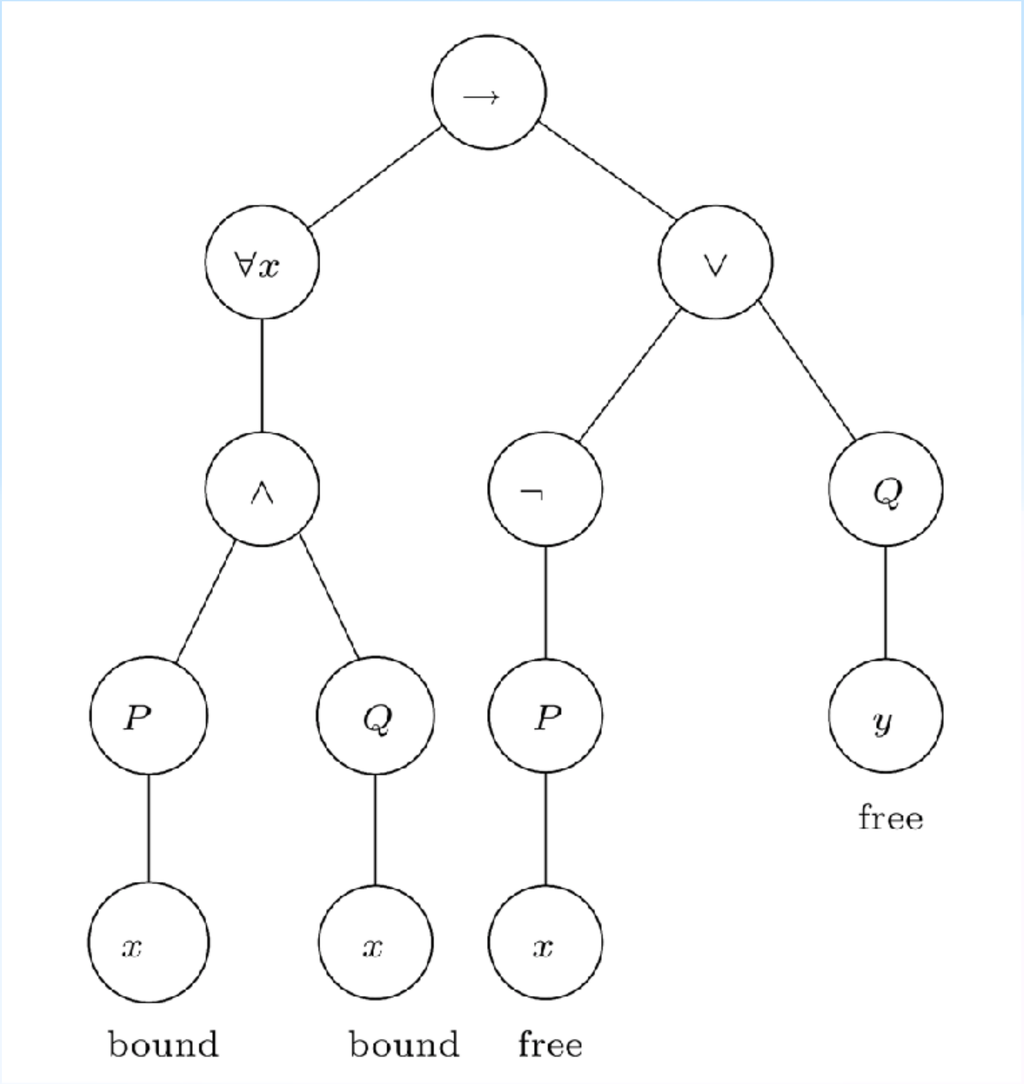


Figure 2.1. A parse tree of a predicate logic formula.

Definition 2.6 Let ϕ be a formula in predicate logic. An occurrence of x in ϕ is free in ϕ if it is a leaf node in the parse tree of ϕ such that there is no path upwards from that node x to a node $\forall x$ or $\exists x$. Otherwise, that occurrence of x is called bound. For $\forall x\,\phi$, or $\exists x\,\phi$, we say that ϕ – minus any of ϕ's subformulas $\exists x\,\psi$, or $\forall x\,\psi$ – is the scope of $\forall x$, respectively $\exists x$.

$$(\forall x\,(P(x) \wedge Q(x))) \rightarrow (\neg P(x) \vee Q(y))$$



Ambiguity

- Every man loves some woman.
- What's the meaning?
- (a) $\forall x\,(M(x) \rightarrow \exists y\,(W(y) \wedge L(x, y)))$
- (b) $\exists y\,(W(y) \wedge \forall x\,(M(x) \rightarrow L(x, y)))$
- Why?

Excercise

Write these as formulas.

1. No student failed the exam.
2. Every lock has a key that opens it. And: there is a key that opens every lock.
3. Everyone's best friend trusts them.
4. Nobody's mother is younger than them.

Say what each of your predicate and function symbols means before you use it.

Solution

1. $\neg\exists x\,(S(x) \wedge F(x))$ or $\forall x\,(S(x) \rightarrow \neg F(x))$ Both are right, and they are equivalent. Which did you write?
2. $\forall x\,(L(x) \rightarrow \exists y\,(K(y) \wedge O(y,x)))$ $\exists y\,(K(y) \wedge \forall x\,(L(x) \rightarrow O(y,x)))$ The second implies the first. Not the other way round.
3. $\forall x\ T(b(x), x)$
4. $\neg\exists x\ Y(m(x), x)$ or $\forall x\ \neg Y(m(x), x)$

Restricted quantifiers — why ∀ takes → and ∃ takes ∧
The awkwardness solved when talking about GQT in week 8

RESTRICTED QUANTIFIERS

English quantifies over **students**. Logic quantifies over **everything**.
So something has to do the restricting — and which connective is **forced**.

GET IT RIGHT

Every student passed	$\forall x\,(S(x) \rightarrow P(x))$	*→ is true whenever S(x) is false — it lets the non-students off*
Some student passed	$\exists x\,(S(x) \wedge P(x))$	*∧ forces your witness to be both — a student, and one who passed*

SWAP THEM, AND…

$\forall x\,(S(x) \wedge P(x))$	***everything** in the universe is a student, and everything passed*	true in **1** of 64
$\exists x\,(S(x) \rightarrow P(x))$	*something is such that if it is a student it passed — and any non-student already makes that true*	true in **63** of 64

They fail in **opposite directions**: the first is almost never true, the second almost never false.
counted over all 64 models on a three-element domain

Solve exercise 2.1.3 (p. 157)

What is a free/bound variable?

Excercise

For each occurrence of each variable, is it free or bound by which quantifier?

1. $(\forall x\ (P(x) \rightarrow Q(y))) \wedge (\exists y\ R(x, y))$
2. $\exists x\ (P(x) \wedge \forall x\ Q(x))$
3. $\forall y\ (P(x) \rightarrow \exists z\ Q(z, f(x, y)))$
4. $\forall x\ \exists y\ (P(y) \vee Q(z))$

Draw the parse tree if it helps. It will.

Solution

1. x in P(x) — bound, by ∀x · y in Q(y) — free x in R(x,y) — free · y in R(x,y) — bound, by ∃y Both variables occur free and bound in the same formula.

2. x in P(x) — bound by the outer ∃x · x in Q(x) — bound by the inner ∀x Nothing is free. The inner quantifier shadows the outer one.

3. x in P(x) — free · z — bound by ∃z x in f(x,y) — free · y in f(x,y) — bound by ∀y Free variables of the formula: x only.

4. y — bound by ∃y · z — free ∀x binds nothing at all. It is still a legal formula.

Definition 2.7 Given a variable x, a term t and a formula ϕ we define $\phi[t/x]$ to be the formula obtained by replacing each free occurrence of variable x in ϕ with t.

SUBSTITUTION — DEFINITION 2.7

Given a variable x, a term t and a formula φ, we define $\boldsymbol{\varphi[t/x]}$ to be the formula obtained by replacing **each free occurrence** of x in φ with t.

SUBSTITUTING THE TERM **f(x,y)** FOR **x**:

$\forall x\,((P(x) \to Q(x)) \wedge S(x,y)) \longrightarrow \forall x\,((P(x) \to Q(x)) \wedge S(x,y))$

*no **free** x anywhere — nothing is replaced*

$(\forall x\,(P(x) \wedge Q(x))) \to (\neg P(x) \vee Q(y)) \longrightarrow (\forall x\,(P(x) \wedge Q(x))) \to (\neg P(f(x,y)) \vee Q(y))$

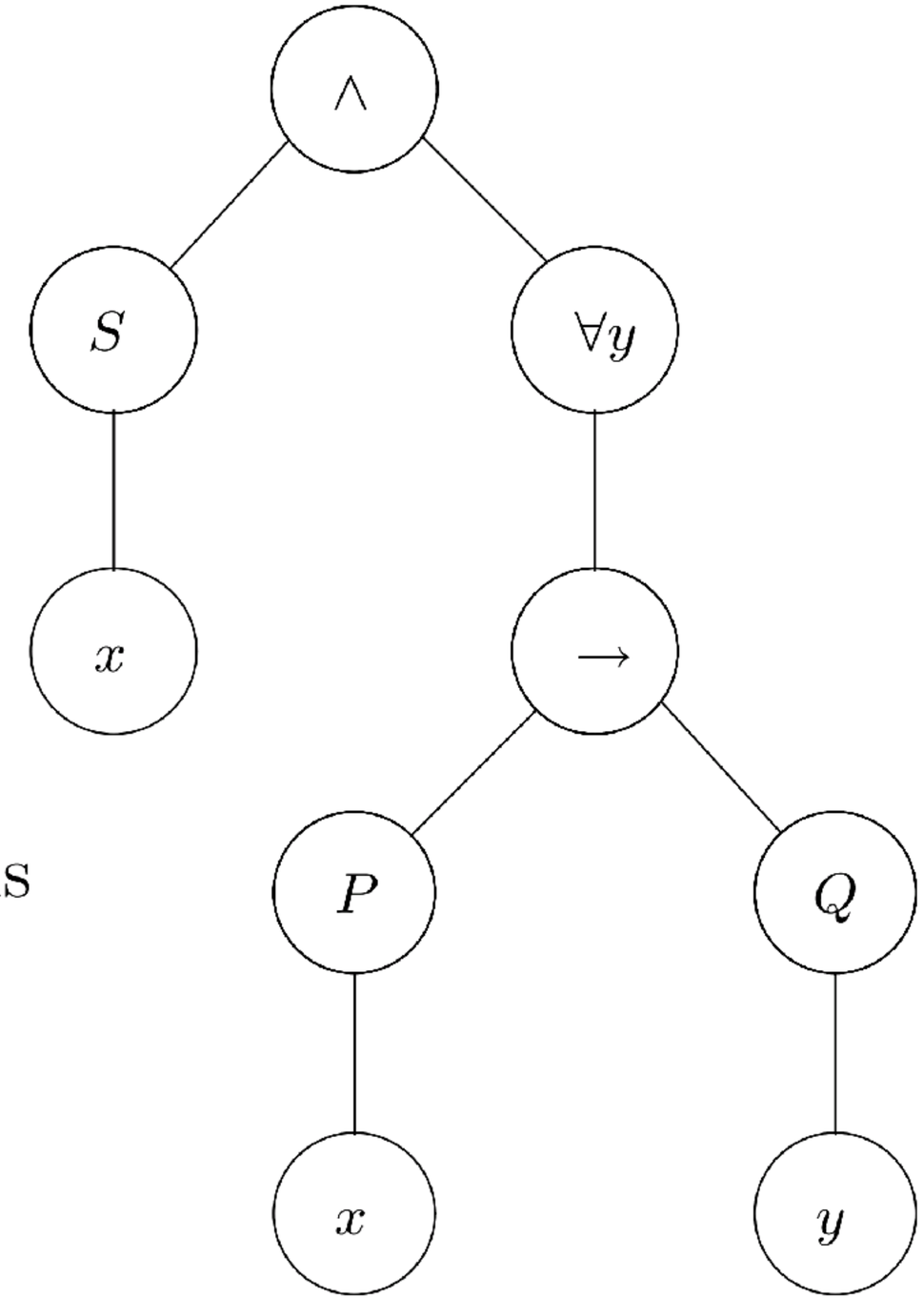
exactly one free x — and only that one changes

“Free for” - concept

- $\phi(x) = \exists y\ L(x, y)$ "x loves someone"
- Substitute y for x:
- $\phi[y/x] = \exists y\ L(y, y)$ "someone loves themselves"
- These say completely different things. The substitution was legal by Definition 2.7 — and it destroyed the meaning.
- Definition 2.8 exists to forbid exactly this.

Definition 2.8 Given a term t, a variable x and a formula ϕ, we say that t is free for x in ϕ if no free x leaf in ϕ occurs in the scope of $\forall y$ or $\exists y$ for any variable y occurring in t.

the term $f(y, y)$ is not free for x in this formula



Excercise

Compute each substitution. Then say whether it was legitimate.

1. $\phi = \forall x\,(P(x) \rightarrow Q(y))$ compute $\phi[g(y)/x]$
2. $\phi = P(x) \wedge \exists y\, R(x, y)$ compute $\phi[f(z)/x]$
3. $\phi = P(x) \wedge \exists y\, R(x, y)$ compute $\phi[f(y)/x]$

Definition 2.7 always gives you an answer. Definition 2.8 tells you whether to trust it.

Solutions

1. $\phi[g(y)/x] = \forall x\,(P(x) \rightarrow Q(y))$ — unchanged There is no free x. Nothing to replace. Vacuously legitimate.

2. $\phi[f(z)/x] = P(f(z)) \wedge \exists y\, R(f(z), y)$ f(z) is free for x in ϕ. Nothing binds z. Legitimate.

3. $\phi[f(y)/x] = P(f(y)) \wedge \exists y\, R(f(y), y)$ f(y) is not free for x in ϕ. The second occurrence has been captured by ∃y. Definition 2.7 computed it anyway. Definition 2.8 is why we do not use it.

Solve exercise 2.2.4 (p. 160)

Hintikka’s sentence:

Some relative of each villager and some relative of each townsman hate each other.

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Branching Quantification v. Two-way Quantification

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Abstract

We discuss the thesis formulated by Hintikka (1973) that certain natural language sentences require non-linear quantification to express their meaning. We investigate sentences with combinations of quantifiers similar to Hintikka's examples and propose a novel alternative reading expressible by linear formulae. This interpretation is based on linguistic and logical observations. We report on our experiments showing that people tend to interpret sentences similar to Hintikka sentence in a way consistent with our interpretation.

If Hintikka is right, you cannot write in the language you just learned — no matter how you nest the quantifiers. Week 8 is about exactly which sentences those are, and, time permitting, I will show you our experiments on whether people actually read them that way.